SANSKAR PHOUGAT

Software Engineer | Backend, Full-Stack & Secure Systems
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SUMMARY

Software engineering student (B.Tech ECE, 2027) who builds backend and full-stack systems in Python, Java and JavaScript. Hands-on with REST and GraphQL APIs, PostgreSQL and MongoDB, Docker, AWS, CI/CD and automated testing, with a research internship in API and system security at CyberPeace Foundation. Instructor to 100+ students.

EDUCATION

Bachelor of Technology, Electronics and Communication Engineering | Aug 2023 – Expected Jun 2027
Jaypee Institute of Information Technology (JIIT), Noida
Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design fundamentals

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL
Backend & APIs: Flask, Spring Boot, Node.js, Express.js, FastAPI, REST API design, GraphQL, Apache Kafka, Microservices
Frontend: React, Next.js, HTML5, CSS3, Tailwind CSS, Responsive Design
Databases: PostgreSQL, MongoDB, MySQL, Schema Design, Indexing, Query Optimization
Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, Bedrock), Docker, GitHub Actions CI/CD, Linux, Bash, Git
Testing & Quality: pytest, JUnit 5, Mockito, Postman, Unit and Integration Testing, Code Review
Security: OWASP Top 10, API Security, Vulnerability Assessment, AES-256-GCM, RSA-4096, Argon2id, TLS, Wireshark, Burp Suite
Core CS & Practices: Data Structures & Algorithms, OOP, Git workflows, Agile-style collaboration, AI-assisted development (Claude Code)

PROFESSIONAL EXPERIENCE

Research Intern, CyberPeace Foundation, New Delhi | May 2026 – Jul 2026

- Reverse-engineered API implementations and system architectures across 5+ production systems; documented 12+ security issues with reproduction detail under responsible-disclosure practice.
- Worked with senior researchers across 3 client engagements on infrastructure-hardening recommendations and structured technical reports.
- Built and tested 2 internal tools: Clarifact (AI fake-news detection browser extension on Node.js/Express and AWS Bedrock) and a memorable password generator; selected through a technical onboarding assessment (MITM attack simulation).

PROJECTS

MedVault: Zero-Trust Medical Data Platform | 2025 – 2026 | github.com/Sanskar-bot/medical-data-decentralisation
Python, Flask, PostgreSQL, MongoDB, Docker, AWS EC2, GitHub Actions, pytest

- Designed a multi-tenant REST API enforcing role-based access control across 4 user roles at both API and database layers, with client-side hybrid RSA-4096 + AES-256-GCM encryption so the server never sees plaintext records.
- Cut query latency by 42% through query-plan analysis, index design and connection pooling; wrote a 40+ test pytest suite. Deployed on AWS EC2 with Docker, GitHub Actions CI/CD, and self-configured TLS and DNS.
- Audited the codebase and fixed 3 endpoints missing authorization checks; raised FHIR R4 compliance from 41% to 68%. Security architecture submitted to ICNSBT 2026.

VaultZero: Zero-Knowledge Password Manager | 2024 – 2025 | github.com/Sanskar-bot/VaultZero
Python, JavaScript, TypeScript, REST API, GraphQL, Argon2id, AES-256-GCM

- Built a unified REST API with 100+ endpoints backing browser-extension and mobile clients; implemented Argon2id key derivation and reusable zero-knowledge encryption libraries.
- Added a GraphQL layer on top of the REST service contracts and tested it separately; wrote TypeScript-based test automation. Live demo: passguard-two.vercel.app

Clarifact: AI Fact-Checking Chrome Extension | 2026 | github.com/Sanskar-bot/clarifact
JavaScript, Node.js, Express.js, Chrome Extension (Manifest V3), AWS Bedrock, REST API, Render

- Built an Express.js backend and Manifest V3 extension that fact-checks selected web text in real time, combining web search with an LLM (Nemotron Nano 3 30B via AWS Bedrock) and showing the verdict in an injected sidebar.
- Validated the integration across 1,000+ news and fact-based test cases, refining input/output formats on edge cases; deployed live on Render.

LEADERSHIP

Technical Head, AI Tronics Society, JIIT Noida | Aug 2023 – Present

- Taught Python, Git and DSA workshops to 100+ students; designed and led a 2-day Applied Cryptography workshop for 40+ students; ran 3 networking workshops; mentored 8+ peer projects on system design.

CERTIFICATIONS AND ACHIEVEMENTS

- TryHackMe: top 15% globally across 40+ labs (tryhackme.com/p/Sanskar2003).
- Google Cybersecurity Certificate (2024); AWS Cloud Practitioner Essentials; Salesforce Trailhead Hiker.
- Merged pull request to SigmaHQ/sigma; DSA practice on LeetCode and Codeforces; CTF participant.